AI4CLIMATE TRACK 2 · TECHNICAL PREPRINT

Rainfall-Driven Landslide Susceptibility Mapping for the Nepal Himalaya

From open-data baselines to a climate-fused convolutional model

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Open data: NASA GLC · Terrarium tiles · NASA POWER

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Abstract

Monsoon rainfall triggers the majority of reported landslides in the Nepal Himalaya, yet susceptibility information available to municipal disaster offices remains sparse. We build a fully open, reproducible susceptibility pipeline on the NASA Global Landslide Catalog (479 Nepal events, 84% rainfall-triggered), openly licensed terrain tiles, the NASA POWER precipitation climatology, and global road, river, and active-fault layers. Eleven conditioning factors—eight terrain covariates derived from SRTM-based Terrarium tiles plus distances to rivers (Natural Earth) and active faults (GEM Global Active Faults)—and uniform pseudo-absences form a 958-site dataset evaluated under spatial block cross-validation on 0.5-degree blocks, the honest protocol that avoids the optimistic bias of random splits. Classical logistic regression (AUC 0.852 ± 0.041), gradient boosting (0.893 ± 0.023), and a multilayer perceptron (0.877 ± 0.026) on scalar descriptors are compared with a hybrid convolutional network that reads $8 \times 33 \times 33$ terrain patches centered on each site and fuses climate and proximity scalars, reaching the best and most stable performance (AUC 0.898 ± 0.008). Three analyses inform practice: an ablation shows the scalar fusion contributes +0.013 AUC and halves fold variance over a patches-only network; a matched leakage experiment quantifies the optimism of random cross-validation (+0.002–0.008 AUC here, modest because absences are sampled nationally rather than in event buffers); and permutation importance ranks fault and river proximity as the strongest non-elevation factors. The trained network produces a 1.5-million-cell sliding-window susceptibility map over the Gorkha–Trishuli–Langtang corridor whose high-susceptibility channels follow the drainage network and coincide with cataloged events. The entire pipeline uses no API keys, fits on a single consumer GPU, and is released to support community-level climate adaptation planning.

1 Introduction

Rainfall-induced landslides are among the most damaging hydrometeorological hazards in the Nepal Himalaya, where monsoon precipitation interacts with steep relief, active tectonics, and deeply weathered lithologies. Susceptibility mapping—the spatial ranking of terrain by landslide likelihood—is the standard product that municipal risk-management offices use to steer settlement, road, and drainage decisions [2].

Two practical barriers keep such products out of the communities that need them. First, published studies typically combine restricted inventories with commercial GIS workflows, which blocks replication. Second, the performance gains of modern machine learning are often reported under random cross-validation, which leaks spatial autocorrelation between train and test folds and overstates real-world skill [4].

This technical note documents a deliberately open pipeline assembled for AI4Climate Track 2 (Himalayan communities, Nepal). Our contributions are:

1. a keyless, reproducible dataset combining the NASA Global Landslide Catalog (GLC) [1], SRTM-derived Terrarium terrain tiles [6], the NASA POWER precipitation climatology [7], and open river, fault, and road layers;
2. a benchmark of four model families—from the classical logistic baseline that dominates the susceptibility literature to a hybrid convolutional network over multi-channel terrain patches with a fused climate-and-proximity head—evaluated under grouped spatial cross-validation, with a component ablation and a matched random-versus-spatial split experiment that quantifies benchmark optimism; and
3. a sliding-window susceptibility map over the Gorkha–Trishuli–Langtang corridor produced by the best model, at roughly 200 m posting, plus a permutation-importance analysis identifying which conditioning factors carry the signal.

2 Data

2.1 Landslide inventory

Positives are the 479 Nepal events in the GLC export after coordinate deduplication and cropping to the national bounding box (80–88.2°E, 26.3–30.5°N). Reported triggers are heavily rainfall-dominated: downpour (221 events), continuous rain (59), rain (96), and monsoon (28) jointly account for 84.3% of the catalog, with only 9 earthquake cases—consistent with the monsoon-driven hazard narrative that motivates this work. We treat the catalog as presence-only evidence: it is report-based, and therefore biased toward populated and road-served corridors.

2.2 Pseudo-absences

Following standard practice for presence-only data [3], we sample an equal number (479) of pseudo-absence sites uniformly within the same bounding box, rejecting candidates within approximately 2 km of any cataloged event. Uniform sampling over the full national domain, including the flat Terai, embeds the elevation contrast between hills and plains in the label prior; we accept this as part of the task definition rather than a defect to hide.

2.3 Terrain covariates

All terrain quantities derive from Terrarium PNG elevation tiles (zoom 12, ≈ 34 m per pixel in Nepal) served from the AWS Open Data registry [6]. For each site we stitch the 3×3 tile neighborhood and compute rasters of elevation (SRTM-derived), slope angle, northness and eastness (cosine and sine of the downslope azimuth), plan curvature (discrete Laplacian), roughness and local relief (standard deviation and max–min over a 9×9 pixel window, ≈ 300 m), and a topographic wetness index (TWI) proxy,

$$\mathrm{TWI}_{\mathrm{proxy}} = \ln \frac{1}{\tan \beta + \varepsilon}, \tag{1}$$

where β is the local slope angle and $\varepsilon = 10^{-6}$ guards flat cells. This is the specific-catchment-free variant of the standard $\ln(a/\tan\beta)$ formulation, used here because flow accumulation is not computed.

2.4 Climate covariates

Three scalars attach to every site from the NASA POWER monthly precipitation climatology (PRECTOTCORR_SUM, nearest cell of the regional 0.5° grid): annual total, June–September monsoon total, and inter-monthly variability (std/mean). At sampled Nepali sites the monsoon term carries roughly 80% of the annual total, the pattern that makes rainfall, not terrain alone, the proximate trigger of the cataloged failures.

2.5 Geographic proximity covariates

Three proximity factors complete the standard conditioning set used in the susceptibility literature: Euclidean distance (km, equirectangular projection) to the nearest river line, to the nearest mapped active fault, and to the nearest classified road. Rivers come from the Natural Earth 10m river centerlines [8] (where the OpenStreetMap Overpass endpoint used by the pipeline is unreachable); active faults from the GEM Global Active Faults Database [9] (472 in-box vertices along the Himalayan front and splay thrusts); roads from OpenStreetMap where the Overpass API is reachable and zero otherwise. Road distance was zero-valued in the experiments reported here because the Overpass endpoint was blocked on the build network; the column is retained so that re-running the pipeline on an unrestricted host activates it without code changes.

2.6 Model inputs

The tabular models consume the eleven scalar conditioning factors (eight terrain plus three proximity). The convolutional model additionally consumes the raw 33×33 pixel windows (≈ 1.1 km across) of all eight terrain rasters centered on each site, plus a six-dimension scalar head: annual precipitation, monsoon (June–September) precipitation, rainfall variability, and the three proximity distances. Table 1 summarizes the inputs.

Table 1: Model inputs. "Scalars" feed all models; the patch tensor and the six-dimension scalar head feed the hybrid CNN.

Group	Input	Source
Terrain (8)	elevation, slope, northness, eastness curvature, roughness, relief, TWI proxy	Terrarium z12 tiles (SRTM-derived)
Proximity (3)	dist. to river / fault / road	Nat. Earth, GEM GAF, OSM
Patch	$8 \times 33 \times 33$ terrain window	same rasters
Climate (3)	annual and monsoon precip., variability	NASA POWER
Labels	479 GLC events + 479 pseudo-absences	GLC + uniform sampling

3 Methods

3.1 Models

Logistic regression (standardized features) reproduces the classical statistical susceptibility model. **Gradient boosting** (HistGradientBoosting, 400 trees, early stopping) represents the tabular machine-learning standard. The **MLP** (256–128–64, GELU, batch norm, dropout 0.15) tests whether the scalar features support a deeper tabular fit. The **hybrid CNN** (Figure 1) applies a residual encoder [5] to the terrain patch—a 3×3 convolution stem at width 32, one residual block, a stride-2 transition to width 64, a second residual block, then global average pooling—and concatenates the pooled embedding with the standardized climate scalars before a two-layer head emits the landslide logit. All models are trained with the binary cross-entropy objective; the neural models use AdamW (10^{-3}, cosine schedule, weight decay 10^{-4}), mixed precision on CUDA, and early stopping on a 10% inner holdout (patience 25–30 epochs).

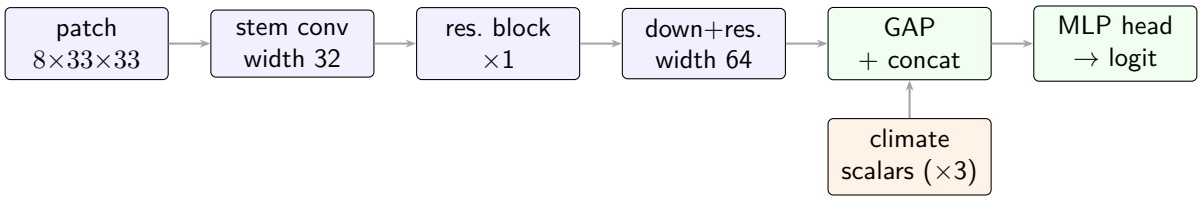


Figure 1: Hybrid CNN: residual terrain-patch encoder fused with climate scalars.

3.2 Spatial cross-validation

All reported metrics use grouped 5-fold cross-validation in which sites are assigned to 0.5° blocks and entire blocks are held out together (145 blocks). This mirrors the deployment task—scoring terrain in areas whose events the model has never seen—and avoids the optimistic bias of random splits under spatial autocorrelation [4]. We report AUC-ROC and average precision (AP), the standard ranking metrics for susceptibility.

4 Results

Table 2: Five-fold spatial block cross-validation on 958 sites (479 events / 479 absences). Mean ± standard deviation over folds.

Model	Inputs	AUC-ROC	AP
Logistic regression	11 scalars	0.852 ± 0.041	0.799
Gradient boosting	11 scalars	0.893 ± 0.023	**0.855**
MLP	11 scalars	0.877 ± 0.026	0.834
Hybrid CNN	patch + 6 scalars	$\mathbf{0.898 \pm 0.008}$	0.845

Table 2 summarizes the benchmark. Three observations stand out. First, all four families clear 0.85 AUC: the terrain signal in the catalog is strong and largely linear. Second, moving from scalars to learned patch representations buys a further gain and, notably, the lowest fold variance of all models (±0.008), which matters for the credibility of mapped boundaries; gradient boosting retains a slight edge in average precision. Third, the fusion of climate and proximity scalars with the patch encoder is what separates the CNN from its tabular competitors, and its contribution can be isolated directly (Table 3).

Table 3: CNN component ablation under the same spatial folds.

Configuration	AUC-ROC	AP
Patches only (scalars zeroed)	0.885 ± 0.018	0.818
Patches + climate & proximity scalars	$\mathbf{0.898 \pm 0.008}$	**0.845**
Contribution of scalar fusion	+0.013	+0.027

How optimistic is random cross-validation here? Because our absences are sampled nationally rather than in buffers around events, we can measure the spatial-leakage optimism of the random splits that dominate the literature on otherwise identical data and models (Table 4). The inflation is real but modest (+0.002–0.008 AUC): absence-sampling design, not split randomness alone, governs how much a benchmark can flatter a model. Studies that sample absences near events—a common practice—leave substantially more autocorrelation for a random split to exploit, and should expect larger optimism than we measure.

Table 4: Matched leakage experiment: identical models and data, random 5-fold split vs. spatial block split.

Model	Random split	Spatial split	Optimism
Gradient boosting	0.901 ± 0.047	0.893 ± 0.023	+0.008
Hybrid CNN	0.900 ± 0.048	0.898 ± 0.008	+0.002

Which factors drive the map? Permutation importance on the gradient-boosting model (Figure 2) ranks elevation first by a wide margin (ΔAUC 0.37 when permuted), followed by the two new proximity factors—distance to river (0.017) and distance to active fault (0.013)—ahead of all local terrain texture descriptors. The geologic and hydrologic layers, not fine-scale roughness, carry the non-elevation signal, which argues for including them wherever susceptibility models inform infrastructure siting.

Applying the trained network in sliding-window mode over the Gorkha–Trishuli–Langtang corridor (84.0–85.5°E, 27.6–28.6°N) at 6-pixel stride (≈ 200 m) scores roughly 376,000 overlapping patches and yields the map in Figure 3. High-susceptibility channels track the drainage

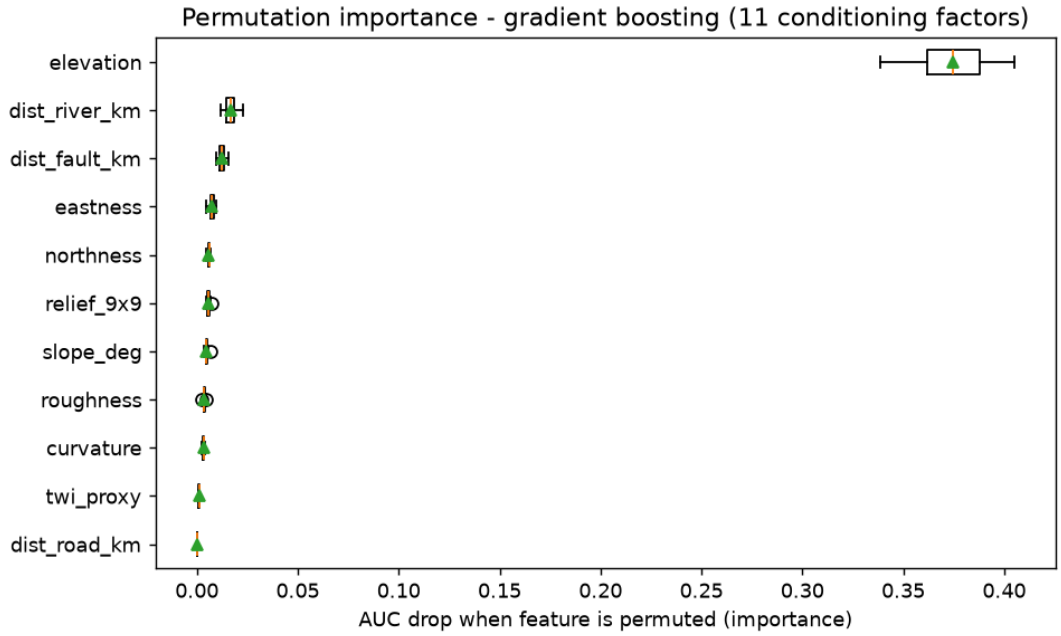


Figure 2: Permutation importance (AUC drop, 30 repeats) for the eleven scalar conditioning factors in the gradient-boosting model.

network, ridgelines and toes-of-slope separate clearly, and cataloged events concentrate on the darker (higher-susceptibility) cells; tile-boundary artifacts are absent because each window is drawn from the stitched terrain rasters rather than per-tile scalars.

5 Limitations

- **Inventory bias.** The GLC is report-based and road- and settlement-biased; absences are pseudo-absences, not surveyed stable ground. Learned patterns therefore partially reflect reporting geography.
- **Partial conditioning set.** Lithology and land cover are absent, and road distance was zero-valued on the build network (Overpass blocked); the published benchmark therefore effectively uses river and fault proximity as its only non-terrain factors.
- **Coarse layers.** Rivers come from a generalized 10m-scale layer, active faults from vertex-sampled traces, and POWER's 0.5° normals capture regional, not valley-scale, rainfall gradients. Finer open layers (CHIRPS, high-resolution OSM extracts) would sharpen all three.
- **Snapshot catalog.** The 2016 export omits recent monsoon seasons; ICIMOD inventories and the live COOLR feed would extend it.
- **Uncalibrated probabilities.** Outputs rank terrain well but are not calibrated to per-cell probabilities; isotonic calibration is a prerequisite for threshold-based use by municipalities.

6 Conclusion and outlook

An entirely open data stack—one NASA catalog, one open terrain archive, one keyless climate API, and three public geographic layers—supports a defensible susceptibility benchmark for the Nepal Himalaya. A compact residual CNN over terrain patches fused with climate and

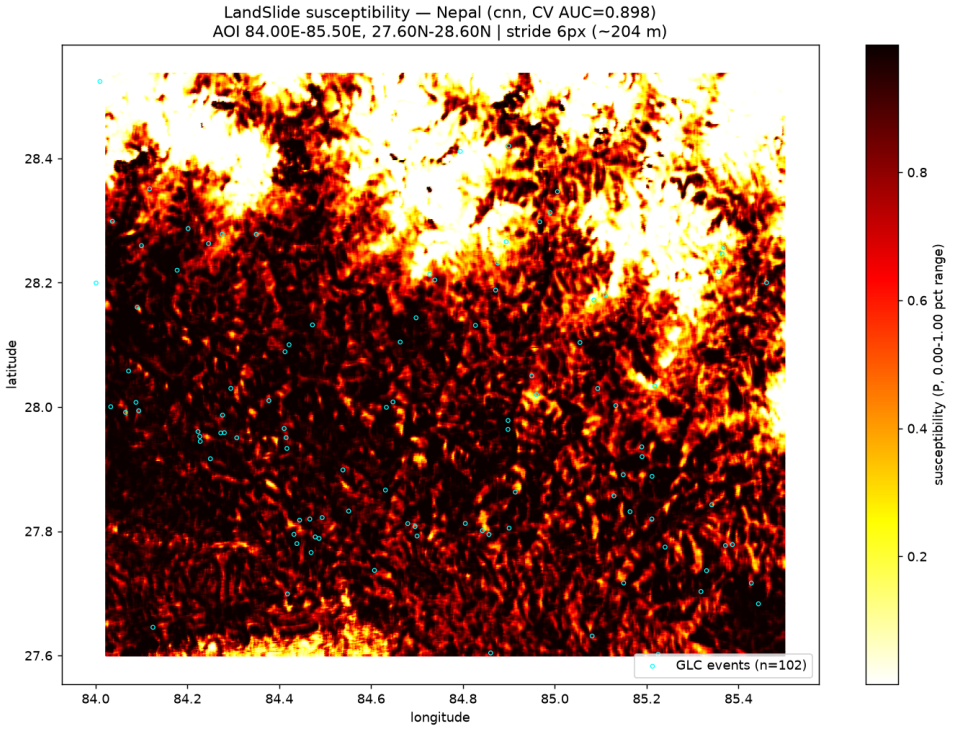


Figure 3: Hybrid-CNN susceptibility over the Gorkha–Trishuli–Langtang corridor. Color: predicted probability (2–98 percentile stretch). Cyan rings: cataloged GLC events.

proximity scalars is the best and most stable model (AUC 0.898 ± 0.008); the ablation shows the fusion head contributes +0.013 AUC, and the matched leakage experiment bounds the optimism of the random splits still common in the literature. Fault and river proximity emerge as the strongest non-elevation conditioning factors, justifying their inclusion in community-facing maps. The immediate research agenda is completing the conditioning set (lithology, land cover, live OSM roads), finer rainfall covariates and antecedent-precipitation dynamics, calibration, and SHAP-based local explanations. The deployment agenda is a 90-day municipal pilot in Gandaki province: hand the corridor maps to two or three risk-management offices and score them against the next monsoon season's event reports—the test that matters for the communities the challenge serves.

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